

PART 2/2

HOTDOCX & JSCOQ

AN INTERACTIVE, AI-AUGMENTED, AND MONETIZABLE
PLATFORM FOR COQ

THE CHALLENGE: SHARING COQ DEVELOPMENTS

Coq is a cornerstone of formal methods, but sharing its rich, interactive proofs is hard.

CURRENT STATE

- Static PDFs (Flattened)
- Code Repositories



THE COMMUNITY'S DESIRE

- Dynamic & Accessible Platforms

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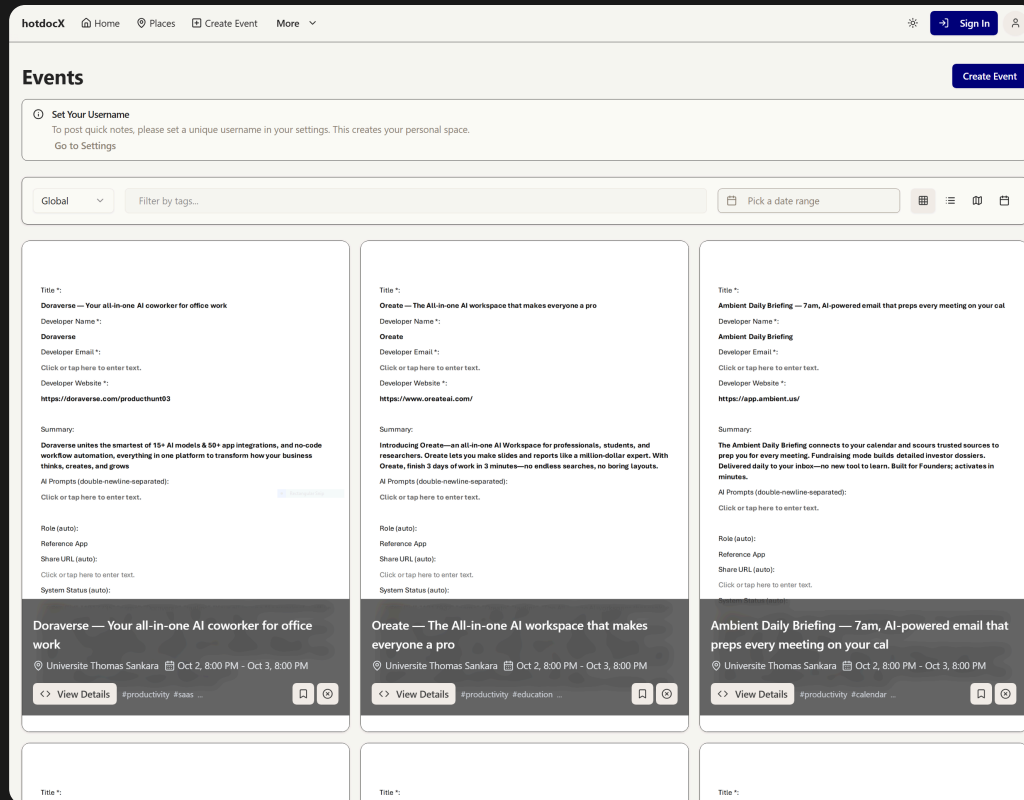
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THE CORE PROBLEM: A MISSING PLATFORM THAT IS BOTH INTERACTIVE AND SUSTAINABLE.

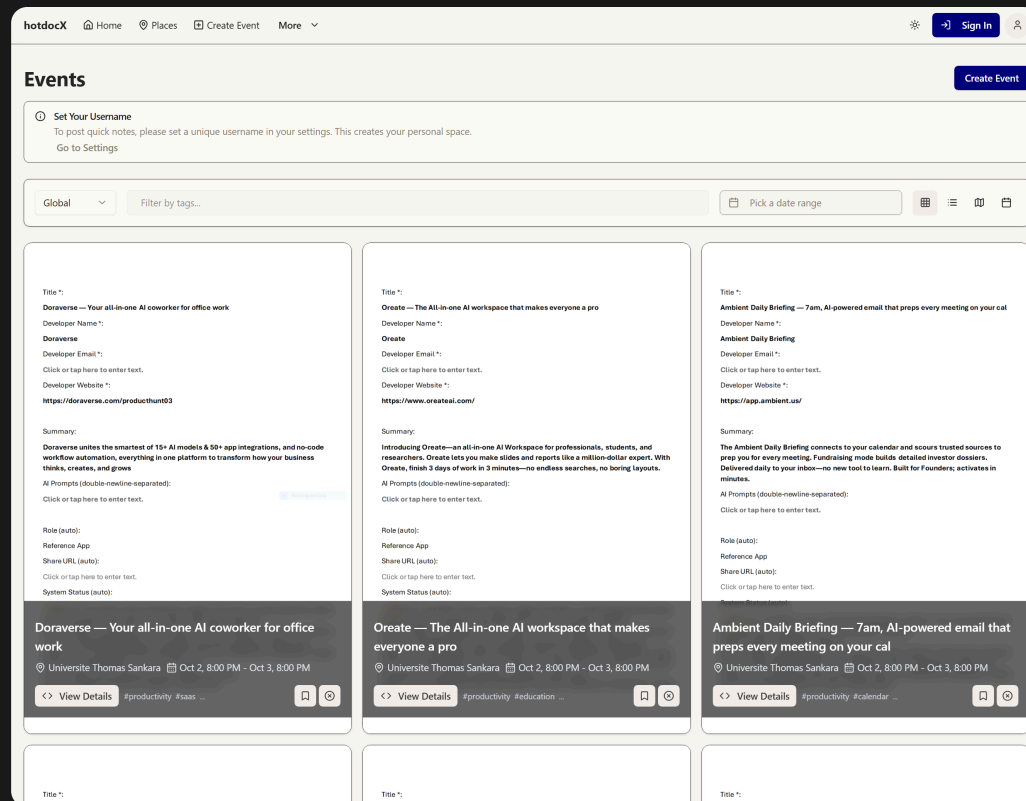
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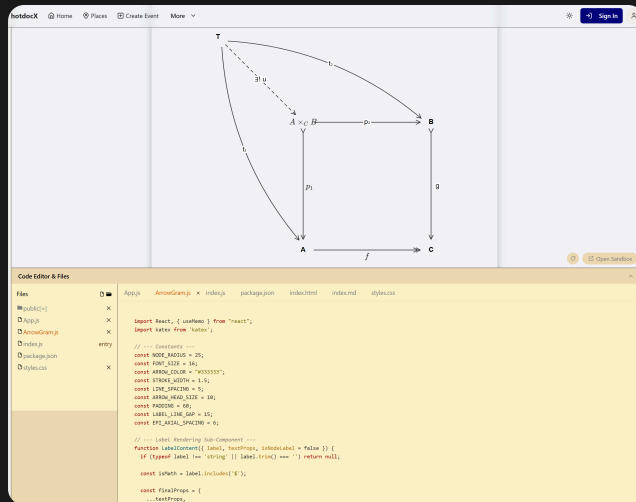


We're moving from "papers-with-code" to "papers-as-apps".

CORE CONCEPT: AI TEMPLATES

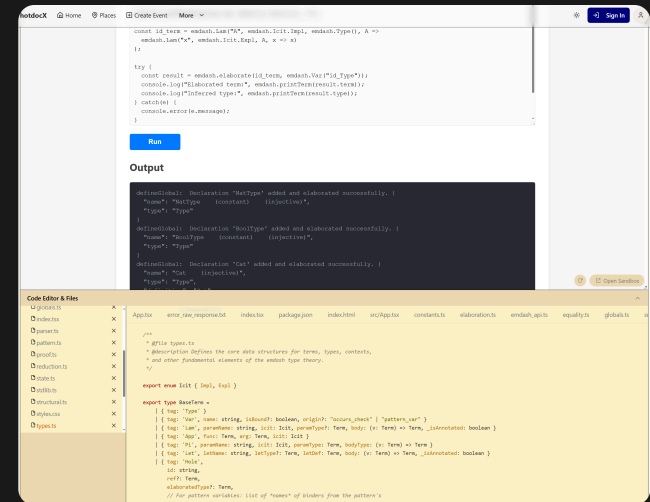
A user provides a document (PDF, LaTeX, etc.) and a prompt. Our AI pipeline uses a template to generate a runnable application.

Arrowgram Template



Renders commutative diagrams from natural language.

Emdash Template

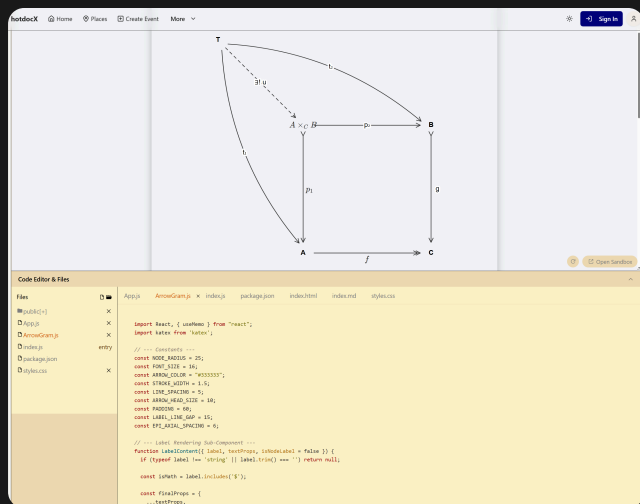


Provides a live environment for our logical framework.

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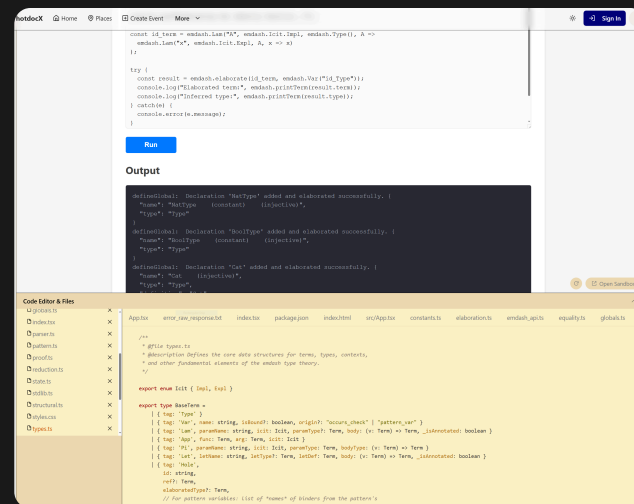
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The platform is flexible and context-aware. Now, let's add Cog to the mix.

MAKING COQ A FIRST-CLASS CITIZEN

We created a ``jsCoq`` template for the hotdocX ecosystem.

Any Coq script can become a first-class, interactive, and shareable hotdocX event.

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LIVE DEMO 1: INTERACTIVE COQ SESSION

hotdocX

HomePlacesCreate EventMore

Sign In

jsCoq Interactive Session

```
1 (* A simple Coq example *)
2 Require Import Arith.
3
4 Check N.
5
6 Theorem plus_n_0 : ∀ n:N, n + 0 = n.
7 Proof.
8   intros n.
9   induction n as [| n' IH].
10  - reflexivity.
11  - simpl. rewrite → IH. reflexivity.
12 Qed.
13
14 Check plus_n_0.
```

Goals

1 goal

n : N

n + 0 = n

Messages

Info

Coq.ZArith.BinIntDef loaded.

Coq.ZArith.BinInt loaded.

Coq.setoid_ring.Ring_polynom loaded.

Coq.Lists.ListTactics loaded.

Coq.ZArith.Zeven loaded.

Coq.ZArith.Zcompare loaded.

Coq.ZArith.Zorder loaded.

Coq.Bool.Sumbool loaded.

Coq.ZArith.ZArith_dec loaded.

Coq.ZArith.Zbool loaded.

Coq.setoid_ring.InitialRing loaded.

Coq.setoid_ring.Ring_tac loaded.

Coq.setoid_ring.Ring_base loaded.

Open Sandbox

Code Editor & Files

Files

index.htmlindex.jspackage.jsonindex.v

public[+]

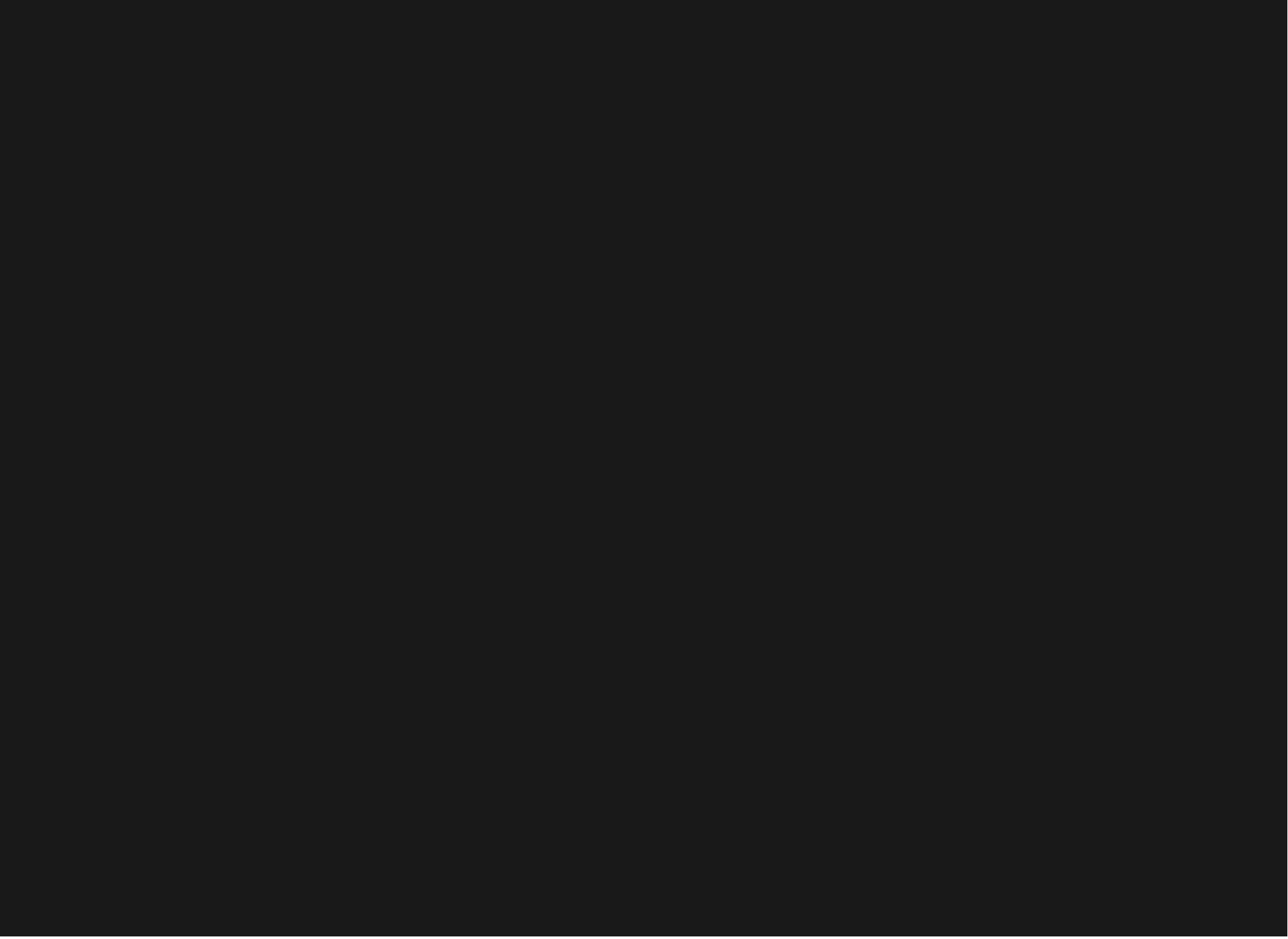
index.html

var jscoq_ids = ['.coq-code'];

var jscoq_opts = {

include: true

An interactive jsCoq session embedded within a hotdocX event.



UNLOCKING POWERFUL NEW WORKFLOWS

EDUCATION

- Create tutorials from lectures + Coq files.
- Students solve exercises in-browser.

RESEARCH

- Publish an arXiv paper and a companion hotdocX event.
- Readers can experiment with lemmas directly.

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- Bridge the gap between paper and formalization.

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- **Monetize Your Content:** Creators define a "Purchase Menu" for events (e.g., view source, remix rights, access solutions).
- **Virtual Currency:** Users purchase "Coins" via Stripe to access premium content.
- **Subscription Tiers:** Free, Premium, and VIP tiers unlock platform features like content mirroring and advanced API access.

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THIS ISN'T JUST A TOOL; IT'S A SELF-SUSTAINING ECOSYSTEM.

THE GRAND VISION: A BROWSER-NATIVE COQ KERNEL

The co-location of `jsCoq` and our native framework, `Emdash`, opens a unique research opportunity.

A HYBRID EMDASH/COQ-LIGHT KERNEL

We can systematically augment the Emdash ($\lambda\Pi$ -calculus) kernel with the features of CIC, all within a native TypeScript environment.

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EDUCATION

A "hackable" kernel for teaching CIC without a complex build setup.

RESEARCH

Rapidly prototype new tactics and language features in TypeScript.

AI INTEGRATION

A native JS kernel is easier for AI agents to interact with than a compiled binary.

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BECOME A SPONSOR

Your contribution, whether individual or institutional, directly funds
development, infrastructure, and community growth.

[GITHUB.COM/SPONSORS/HOTDOCX](https://github.com/sponsors/hotdocx)

Sponsoring provides premium and VIP tiers on the hotdocX platform.

THANK YOU

QUESTIONS & DISCUSSION

Platform: hotdocx.github.io

Sponsor: github.com/sponsors/hotdocx